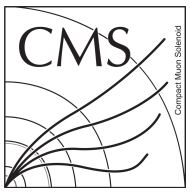
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The Compact Muon Solenoid Experiment
CMS Performance Note



Dual-threshold time-walk calibration for CMS BTL

CMS

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Abstract

With new high intrinsic precision timing detectors for the HL-LHC upgrades, the strategy for commissioning these detectors must differ from previous collider experience. Prior to LHC, collider experiments benefitted from a clear timing reference for use in calibration of the detectors. At low luminosity an interaction trigger was useful for rejecting background but also fulfilled this role. For the ATLAS and CMS upgrades, an interaction trigger would be superfluous since they are designed for 140-200 interactions per trigger event. Dual threshold timing enables a calibration strategy when a t_0 reference is not readily available. We extend an earlier bench test demonstration of this technique with the CMS Barrel Timing Layer(BTL) sensors and the TOFHIR readout ASIC to a 180 GeV pion beam exposure. These data confirm the earlier single coefficient correction for walk and the conclusion that walk calibration to the required $\sim 10\%$ precision is practical with simple data sets using only this system with beam or cosmic rays.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.